

INSTITUTE OF ACTUARIES OF INDIA

CP1B - Actuarial Practice

November 2024 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**a)****1. Market potential –**

- The domestic market may be under-penetrated, despite being a rapidly developing economy, with ample scope for growth in the financial services industry, especially the life insurance market.

2. Capital Availability -

- Galaxy Insurance may have surplus capital to invest and Super Galaxy may deem fit for a high return on capital, considering the scope for growth.

3. Desire to expand into emerging markets

- Galaxy Insurance may desire to enter the emerging market for higher profitability or diversification.

4. Marketing and Distribution Strategies

- Galaxy Insurance may want to implement certain marketing, sales, and distribution strategies in the local market but hasn't been able to do so due to minority shareholding.

5. New partner

- Since the local partner is on the lookout for a buyer, Galaxy Insurance may not be confident and contented in continuing the joint venture with a new leadership.

6. Valuation

- Galaxy Insurance may be getting a good valuation due to the urgency of the local partner to exit.

7. Better Governance

- Galaxy Insurance may practice high standards of good governance including superior internal controls and practices.
- The joint venture may have poor standards at the current state leading to lower growth.

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b)**1. Market Potential and Competition:**

- How big is the market? Growth potential
- Competition in terms of number of players, distribution spread
- Any competitive advantages the company has vis-à-vis the competition

2. Economic Scenario and government policies

- What are the current interest rate levels in the economy?
- Government economic policies? Do they encourage growth and more international investments or very inward-looking policies?
- Is the capital market well developed with appropriate regulatory mechanism
- How are the investor sentiments currently in the economy?

3. Solvency Ratio

- How solvent is the company? Is it operating at a healthy solvency ratio?
- Will the company need any capital injection soon?

4. Expenses

- Current levels of expense ratios
- Expense ratio trends of the company in recent years

- Levels of variable/fixed expenses
5. Market share
 - Market share growth/degrowth tendency
 6. Brand Value
 - How valuable is the brand in the market
 - Is there a need for re-branding? Is any significant investment required for marketing?
 7. Productivity
 - What are the productivity levels of sales staff?
 - Is the company burning more money for additional sales?
 8. Distribution
 - Direct sales level
 - Banca partners
 - Diversification of distribution partners
 - Digitization
 9. Persistency
 - What is the persistency experience of the company?
 - Is there any improvement in the persistency trend? If not, has it been taken into account for future profitability projections?
 10. Mortality/Longevity
 - How is the mortality experience of the company? Have significant death claims been paid out in the recent past?
 - What have been the longevity levels in the demography? Is it improving over the years due to medical advancements and/or improved lifestyle?
 11. Underwriting
 - How strict/lenient is the underwriting process of the company? Are there any gaps in underwriting leading to fraudulent claims being honored and/or higher risk being unidentified?
 - Is the underwriting process advanced enough to identify and categorize risky profiles?
 - How does the company's standards compare with those of the competition
 12. Regulatory Controls
 - What are the levels of Regulatory Controls on the company?
 - Are the regulations encouraging industry development?
 - Are they prescriptive or principle-based
 13. Reinsurance
 - Does the company have acceptable levels of Reinsurance arrangements?
 - Are the reinsurance prices higher for the company? If yes why?
 - Does the company have Stop Loss/Catastrophe reinsurance arrangements in place?
 - Level of technical support reinsurers provide
 14. Product Mix
 - What kind of product mix the company is selling? Is there a healthy mix of all segments of products?
 - Does the product mix have a higher proportion of lower profitability products?

15. Profitability

- What are the profitability levels of the company?
- What have been the profitability trends in the past few years?
- What are the projected levels of profitability of the company?

16. Geographical Concentration

- Is the customer base concentrated in a few states in the country?
- Are the major sales proportions being contributed from specific locations? This makes the company vulnerable to concentration risk due to any catastrophe, competition, etc.

17. Peer Comparison

- How are peers valued compared to the company?
- If there has been an acquisition in the recent past, Galaxy Insurance can refer to the deal to compare the valuation levels

18. Historical and projected sales

- Past trend of sales growth
- Expected future sales volume
- Market potential

19. Assets – Liquidity, Credit Risk, Asset Mix

- What kind of assets the company is currently invested in? Do they carry significant risks?
- Has the company accounted for Liquidity and credit risk in the asset portfolio?
- What are the levels of equity in the portfolio?
- Are the liabilities and assets of the companies matched by duration and cash flow?

20. Customer complaints, litigations, legal risk

- How are the grievance rates/customer complaints levels?
- Are there any litigations on the company?
- Does the company carry any legal risk for the products sold in the past?

21. Checks and controls/ Audit Reports

- Checks and controls in the different functions of the company
- Any system/compliance certifications received
- Past and current audit report findings

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c)

1. Mortality/Longevity

- Past mortality trends of the company.
- Average age group at different lines of business.
- Any demographic characteristic exhibiting higher experience levels.
- Smoker V/s Non-smoker, Gender, Ticket size mix, etc. Are there significant differences in the experience levels under these cohorts?
- What are the levels of reinsurance rates? Is the company retaining acceptable levels of mortality risk?
- Does the company's liability reflect the actual mortality experience at different levels? Are there sufficient levels of prudence considered while determining the policyholder's liabilities?
- Does the mortality assumption used in valuation reflect the above considerations?

2. Persistency

- Historical persistency trends of the company

- Are the persistency levels low for specific products/segments or the overall portfolio?
- Are the persistency levels skewed? Is it very low for some specific channels and higher for the rest?
- What is the experience in early surrender/lapses? If there are a higher number of early lapses, it indicates mis-selling and hence impacts the long-term persistency of the portfolio.
- Does the company's liability reflect the actual mortality experience at different levels?
- Does the company's liability reflect the actual persistency experience at different levels? Are there sufficient levels of prudence considered while determining the policyholder's liabilities?
- Does the persistency assumption used in valuation reflect the above considerations?

3. Expenses

- What are the levels of initial and renewal expenses of the company?
- Are the expenses categorized and allocated in line with their type and nature? – Fixed Vs Variable, Direct Vs Indirect
- Are the future expenses inflated in line with the past trend and future expected inflation levels?
- Does the company's liability reflect the actual expense experience at different levels? Are there sufficient levels of prudence considered while determining the policyholder's liabilities?
- Does the expense assumption used in valuation reflect the above considerations?

4. Interest rates and risk discount rates

- What are the future levels of interest rates assumed? Are these in line with the peers and market expectations?
- Do the interest rates reflect the portfolio's different asset classes?
- Are the liabilities valued at rates consistent with the asset classes it is invested in?
- How sensitive are the liabilities to the interest rate movements?
- Are there sufficient levels of prudence considered while determining the policyholder's liabilities?
- Do the risk discount rates reflect the shareholder's required risk premiums?

(8)

d) It is a strategy with the primary objective of matching the liabilities by duration, term, and value and maximizes the investment return subject to an acceptable risk level as approved by the Board. It is a set of rules, behaviors, or procedures, designed to guide an investor's selection of an investment portfolio. Don't we need to say that?

1. Cashflow matching: Are the asset cash flows closely matched with liability cashflows? If not, what is the level of gap between the two? How has the company planned to manage the mismatch?
2. Credit Risk: What is the credit rating of the assets on the portfolio? If the assets are very risky, it can lead to defaults and hence impact the company's ability to pay claims. Does the company have an investment policy concerning the mix of different asset classes?
3. Duration: Duration of the assets and liabilities need to be closely matched to avoid the disproportionate impact of market movements on asset-liability position.
4. Regulatory and Solvency requirements: Does the investment strategy allow for regulatory and solvency restrictions on asset classes to be invested in?

5. Diversification: Is the portfolio well diversified? What is the proportion of debt and equity in the portfolio? Is the equity investment well diversified across different sectors of the economy?
6. Currency: Does the investment strategy allow for currency risk if any? Are the liabilities and assets matched in the same currency? If not, how has the company managed the currency fluctuation risk?
7. Liquidity: Does the investment strategy manage the liquidity risk of the company? What is the liquidity level the company maintains? What is the proportion of investments in liquid assets such as cash and money market?

(6)

e)

1. Investment risk: The company would be required to manage the investments to provide guaranteed returns to its policyholders. The assets need to be managed such that there is enough buffer for uncertainties and the company is able to generate surplus after providing guaranteed returns to its customers. It faces the risk of long-term guaranteed returns.
 - The monetary policy of the Central Bank and the stance of the government on the current levels of interest rates and the economy will help indicate the expected levels of interest rates.
 - Stochastic modeling can be done to come up with the guarantees that can be offered at a specific Confidence Interval (say 99.5%). A haircut can then be taken on the stochastic guaranteed return to meet the guarantees under an adverse scenario and generate profits.
 - The company can manage the risk by entering into derivative contracts at a higher rate compared to what is guaranteed to the policyholders.
 - While entering into the contract, it is important to check the credibility of the investment bank to avoid counterparty default risk.
2. Expense risk: Since the product is very competitive, it could be priced at thin margins. Hence, the company must manage the expenses (both initial and renewal) effectively. Expenses higher than expected levels can erode the margins and make the product loss leader.
 - Expense risk can be mitigated by continuous monitoring of expenses.
 - It is important that the distribution costs are controlled and maintained at the allowed levels.
 - Looking at cross sales etc., to increase sales on a marginal cost basis besides reducing the unit costs
3. Persistency risk: With high levels of guarantee in place, the surrender value is also likely to be high. Hence, it is likely that the company can incur losses on policy discontinuance. If the company has entered into derivative arrangements, it can further increase the risk of asset mismatching for hedging purposes leading to further losses.
 - The company can manage the risk by controlling the surrender values being offered (especially in the initial durations).
 - Further, persistency-based incentives and/or clawbacks can be introduced concerning distribution partners to manage the persistency levels.
4. Operational risk: Since it's a new product, the company would likely need to set up administration systems to manage the claims and payouts. Even though it's a standard without-profits product, the system has to take care of the guaranteed payouts to the

policyholder along with the management of derivative contracts. It is likely that the admin system can fail and/or not be configured accurately causing losses and other operational hazards for the company.

- The company needs to monitor and put in place appropriate controls if any third-party vendors are involved with the administration
- Rigorous testing with multiple possible use cases would help validate the systems.
- Periodic audit checks would help highlight the key concerns concerning the relevant systems.

5. Solvency risk: Selling high volumes of the product may cause solvency issues for the company, especially if the product has high new business strain. This may require the shareholder to inject further capital to maintain the required solvency levels. Further, market shocks without appropriate hedging mechanisms will also prove to be detrimental to the company's solvency position.

- The company can opt for appropriate reinsurance mechanisms to share risk as well as to get credit in solvency capital calculations. Besides that, it may also explore financial reinsurance to get some relief from new business strains.
- The company can manage the new business product mix as per its risk appetite.
- Regular monitoring and testing of hedge effectiveness would help manage the risk.

(8)

f)

- Funds may be performing well hence declaring healthy bonuses making the product attractive.
- The sales proposition of participating products may be attractive to the target customers.
- Due to investments in riskier assets, the return from participating products may be much better than the guaranteed endowment products.
- The economy could be booming leading to lucrative returns on riskier asset classes.
- The commissions and remuneration on participating products may be high leading to a larger supply and push of participating products from the insurer.

Participating product value proposition:

Policyholder :

- Participating products helps the policyholder to get exposure to riskier asset classes providing better returns. This higher return is managed well through bonus smoothening over the tenure of the policy. Hence, it helps safeguard the returns and interests of the policyholders. – 1 mark
- Unlike unit-linked products, with-profit products have an underlying guarantee as bonuses declared once are guaranteed for life besides the guaranteed SA thus helping the policyholder to get a minimum return.
- The cross-subsidy across generations of policyholders due to the build-up of the estate shall ensure some protection against the volatility of investment returns. Hence, it could be considered to be a better product for policyholders with limited market knowledge.
- A major proportion of the returns from the policyholder fund is shared with the policyholder, whilst the shareholder retains a small fraction of the returns (typically in a 90:10 ratio). Hence, it could be considered to be a lucrative proposition for the policyholders.

Insurer :

- Participating products help the insurer to share the company expenses with the participating policyholders' fund and hence reduce the overall impact of higher expenses.

- The level of commissions and expenses on acquiring and managing the participating portfolio might be comparatively less, leading to its attractiveness.
- Participating products might be relatively more profitable for the insurer.
- Participating in policyholders' undistributed funds might help improve the solvency position of the company (if the regulation allows). This would further add to the attractiveness of the product.
- Participating products might help the insurer to tap a niche market segment and help improve the brand value.
- Participating product might be easy to sell and hence create opportunities for upsell/cross sell other insurance products.

(6)

g) The actuarial control cycle is a fundamental tool of risk management – It involves the process of analyzing, quantifying, mitigating, and monitoring risks. The central part of the model is based on a simple approach to problem-solving:

1. Defining the problem
2. Designing and implementing a solution
3. Monitoring the effectiveness of the solution and revising it if necessary.

In addition, the requirements of professionalism must be recognized at all stages of the cycle.

Actuarial Control Cycle for Financial Management:

1. Specifying the problem:

- Set an aim/objective – for ex., managing the investment risk of the company, controlling the mortality experience, increasing the policyholders' persistency.
- Identify the risks
- Possible mitigation options – All the options available as mitigation strategies.

2. Developing the solution:

- Building a model – Profit testing model/pricing model; deterministic/stochastic model to test and develop a solution. Models can be used to arrive at multiple solutions to a given problem and test for the best option available.
- Assumptions for the model – Multiple assumptions will be required for a functioning model to provide reliable solutions. Assumptions can be demographic and/or economic. Certain approximations and simplifications will need to be taken wherever required to avoid spurious accuracy.
- The actuary may need to decide certain levels of prudence in the assumptions to manage uncertainties.
- Sensitivity testing might be required to understand the credibility of the model. The insight into the relationship between the output and assumptions can be better understood through multiple rounds of sensitivity testing.

3. Monitoring the experience:

- The emerging experience will need to be monitored at regular intervals to comprehend the reliability of the solution.
- The actual and expected parameters can be compared regularly to understand if the expectations are aligned with the experience.
- If the deviations are consistent, the gaps will need to be identified and the expectations will be required to be rebased.

4. Professionalism

- While performing the various tasks under the Actuarial Control Cycle, the Actuary needs to be mindful of compliance with relevant regulations and professional guidance.

5. The general economic and commercial environment

- The Actuary needs to be fully aware of the environment in which the problem needs to be solved.
- The environment relates to both internal and external. The regulatory and commercial environment in which the Actuary operates needs to be fully understood within the actuarial control cycle framework.

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Solution 2:

- a) Third-party insurance is the coverage purchased by an insured member (first party) from an insurance provider (second party) for protection against claims of another (third party). Third-party liability insurance is a type of policy under which the insurance company offers cover for the insured against legal liabilities that arise due to the loss/damage caused by them to a third person's body or property.

The possible reasons for the law enactment:

- Financial Security: Third-party insurance would enable financial security for the insured (first-party) in case he/she becomes liable for damage to the life and/or property of a third party involved in an accident.
- Indemnity and protection of the aggrieved party: Since the third-party insurance indemnifies the aggrieved in case of loss due to an accident, it ensures the financial security of the third party. This would build trust in the legal system and the government.
- Increased cases of Accidents: As the country has high numbers of motor vehicles, accidental cases could be high. This may result in high volumes of crime and law infringement to avoid financial losses. To avoid this, the government may have decided to make third-party insurance mandatory.
- Medical attention to the injured: Without third-party insurance, providing proper medical attention to an injured person at the time of the accident will be challenging.
- Boost the Insurance sector in the country: The government may aim to uplift the economy by providing a boost to the insurance sector. Making third-party insurance coverage mandatory would open new avenues for insurance companies to invest in the country thus directly contributing to the growth of the economy
- Following the developed nations: The government may have taken learnings from the developed nations in terms of what reforms worked best for them. Making third-party insurance mandatory may have proved to be the right step for the other developed nations.

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- b) Stakeholders affected:

- Public/Policyholders
- Motor companies
- Insurance / Reinsurance companies
- Distributors/Brokers/Insurance aggregators
- Government

- Third party administrators (TPA)
- Health care providers
- Public/Policyholders :
 - Government may offer tax rebates on premiums paid which would help reduce the higher cost.
 - Mandatory third-party insurance will provide peace of mind from unforeseen financial obligations.
 - It may result in increased cases of reckless driving.
 - The cost of ownership of a motor vehicle may now increase.
- Motor Companies:
 - Motor companies can bundle the insurance product with the vehicle to help boost profitability in the form of commissions.
 - Profitability can also increase as the companies can now increase the spare parts and servicing price on the back of third-party claims.
 - Higher ownership costs may lead to a fall in demand for the vehicles in the short run.
 - On the other hand, the motor companies can make compulsory insurance as a selling point by explaining the benefits available now that were not there earlier
- Insurance companies:
 - Mandatory third-party insurance will lead to higher sales and hence will increase the profitability of the company.
 - Higher sales may drive higher competition and hence may lead to price wars.
 - Higher sales will lead to efficiencies and help reduce the operating costs of the company.
 - Due to mandatory third-party insurance, there may be a higher incidence of claims resulting in losses.
- Distributors/Brokers/Insurance aggregators:
 - Increased demand for third-party insurance coverage will lead to increased volumes of commissions for distributors. This will increase their profitability.
 - Higher demand may lead to an influx of new distributors causing an increase in competition and resulting in price wars.
 - There may be increased cases of misselling.
- Government:
 - Mandatory third-party insurance may lead to foreign investment in the country which will help boost the industry as well as the economy.
 - Investment in the insurance sector will generate more employment and help the government increase employment levels
 - Taxes levied on commissions will increase the tax collection of the country.
 - The government may grant tax exemptions on premiums paid to increase public confidence and hence tax collection may fall to some extent.

However, increased volumes of premiums should compensate for lower tax rates.

- Third-party administrators:
 - Third-party administrator income will increase on account of higher investments by insurance companies.
 - TPAs can scale up their business and increase the man-power leading to higher employment levels.
 - Higher claim incidences may lead to higher costs for TPAs and hence reduce profitability.
 - TPAs may need to negotiate the fee structure with the insurance companies in such a scenario.
 - The quality of claim investigation may also deteriorate due to higher incidences leading to higher losses for the insurance company.
- Health care providers:
 - Better quality health care can now be provided at higher volumes leading to better profitability.
 - There can be incidences where the insurance company can decline the claims leading to issues for the health care providers to provide appropriate care.
 - The demand for health care may rise leading to expansion of the health care sector which will be beneficial for the general public.
 - Health care providers may increase the prices of their services since the insurance company will now bear the burden of the costs. This may lead to conflict and hence a network list of hospitals may then be approved for treatment of the third party.

(12)

c) Short term:

- Increase in economic growth and employment
- Higher investments leading to progress of multiple sectors including real estate, IT, etc.
- Better knowledge sharing with foreign partners along with the adoption of best practices leading to better services and best products
- Brand differentiation through better quality products and services
- Increase in competition leading to lower prices
- Increase in innovation in the systems and processes with technology transfers
- Better trade relations with other countries

Long term:

- Increased levels of insurance penetration
- Foreign partners may divest if the claim rates do not stabilize
- The industry may stabilize with secular growth rates
- The premium rates may even stabilize at lower levels to avoid the threat of new entrants in the market
- Domestic companies with limited financial resources may not be able to compete with the multinationals over the long term leading to exits
- Exits by insurers will lead to reduced competition and hence, may reduce the quality of services

- Risk of Oligopoly market
- Regulatory reforms to cater to the market needs
- Legislative changes may be introduced to restrict/increase the foreign partner's stake in the insurance sector

(6)

d)

- Higher claims – Higher claims will lead to losses and solvency issues. Hence, maintaining claim rates is crucial.
- Fraud – The company is exposed to the risk of fraud from multiple stakeholders: Policyholder, distributor, health care provider, etc.
- Underwriting – The company is exposed to underwriting risk, especially at the claims stage. Inefficient underwriting will lead to payment for invalid claims and hence impact profitability.
- Reinsurance coverage – There is a risk of insufficient reinsurance coverage available to the company. A reinsurance cover that is not commensurate with the risk appetite can lead to erratic claims payment patterns and solvency issues.
- Reinsurance rates – If the reinsurance rates increase, it will have a direct impact on the company's profitability.
- Claims servicing – If the claims servicing is not optimal, it would lead to customer dissatisfaction and hamper the brand image. Quick resolution of claims and customer queries is essential to maintain customer loyalty.
- Renewal – The insurance company requires repeat customers to maintain a high customer base and increase new business volumes. If the customers do not renew their third-party insurance policies with the company the company will not be able to maintain profitability levels.
- Administration risk – The insurance company will need good levels of administration to manage multiple stakeholders and types of claims. It faces administration risk concerning failed systems and processes.
- Third party risk- If any of the processes are outsourced to third parties, their failure may lead to increased risk for the company.
- Higher expenses: The company faces the risk of high expense levels especially concerning administration and litigation.
- Exposure risk – Lack of diversification will lead to concentrated claims and ineffective monitoring of claim rates.
- New business risk – The company is exposed to new business risk of lower new business sales.
- Competition risk – If the business is profitable, the company is exposed to competition risk with new market entrants creating disruption and eroding the profits.

(5)

e) Possible Reasons

- Policyholder behavior – Policyholders may have become negligent due to the presence of insurance cover.
- Cost of Healthcare – Healthcare costs may have risen resulting in higher claim amounts. Healthcare providers may have been charging higher amounts from insurance companies for medical treatments of third parties.
- Higher repair cost inflation: There may have been an increase in repair and spare parts costs leading to higher claim amounts being paid.
- Geographical spread of policyholders – The policyholders of the company may not be spread enough for diversification. This would expose the company to concentration risk and can lead to higher claims.

- Vehicle type – The company may have a high share of insuring fast vehicles which carry a high risk of accidents and damage. This would expose the company to higher claim rates and amounts.
- Fraud – The insurance company deals with multiple stakeholders: TPA, policyholders, distributors, etc. Higher claims may have been a result of fraud from any of the stakeholders.
- Lower reinsurance coverage – The company may not have adequate reinsurance coverage to share the burden of high claims. This would result in the company bearing the majority or complete burden of claims.
- Inefficient administration – The company may have inefficient administration concerning systems and processes. This can lead to mismanagement, especially in litigation claims leading to losses.
- Claims Underwriting – There may be gaps in claims underwriting leading to honor of invalid claims causing a rise in claim rates.

(3)

f) Actions to be taken :

- a. Stricter claims underwriting
- b. Appropriate model to predict claims reasonably well leading to appropriate pricing
- c. Fraud prevention unit in the company
- d. The geographical spread of business
- e. Using telematics and/or other tools for risk categorization
- f. Appropriate levels of reinsurance to manage losses.
- g. Identification of partners with the highest levels of claims and discontinue business with them
- h. Identify the locations with the highest claim concentration. Stop business there or stricter underwriting
- i. Cost control to manage profitability
- j. Repricing the product to allow for higher expected claims in the future
- k. Reputed third-party administrators to control fraud claims
- l. Blacklisting dubious hospitals

(5)

g)

- To service high levels of claims experience.
- There might be uncertainty over claims experience in the near future.
- Known high amounts of claims might be expected to be serviced.
- Economic situation -The interest rates might be expected to increase in the near future which would make long-term debt unattractive, hence holding cash.
- Other asset classes equity, property, etc. might be overpriced.
- Uncertainty over future economic situation.
- Awaiting/Timing an investment opportunity in the market.
- Maintain the liquidity ratio of the company.

(3)

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